ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** an oilfield	Offshore
2	**(a)** cement making	Calcium
3	**(d)** anthracite	Hard coal
4	**(c)** aluminium	Non-ferrous
5	**(b)** ocean tides	Sea
6	**(b)** organic waste	Cow dung
7	**(a)** metallic mineral	Ferrous
8	**(c)** coal	Fuel
9	**(a)** extraction of minerals from the earth	Industry
10	**(b)** an electrical insulator	Electronic goods

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Biogas is a gas produced from organic waste such as dung (1); It is used for cooking and lighting (1).	1+1
12	Thermal power is electricity generated by burning coal, oil or gas (1); It is produced in thermal plants (1).	1+1
13	Nuclear power is electricity from atomic fuels like uranium and thorium (1); Tarapur is a station (1).	1+1
14	Natural gas is a clean fossil fuel found with petroleum (1); It is used as CNG (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mica (1); Limestone (1); Salt or gypsum (1).	1+1+1
16	Switch off unused lights (1); Use public transport (1); Use efficient appliances (1).	1+1+1
17	Coal (1); Petroleum (1); Natural gas (1).	1+1+1
	OR: Iron ore (1); Copper (1); Bauxite (1).	1+1+1
18	Recycle scrap metals (1); Use substitutes (1); Reduce wastage in mining (1).	1+1+1
	OR: Thermal power generation (1); Industrial fuel (1); Domestic use and coke making (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Minerals occur in veins and lodes in igneous rocks (1); They occur in beds in sedimentary rocks (1); Some are formed by decomposition of rocks (1); Some are found as alluvial deposits (1); Minerals are unevenly distributed (1).	1+1+1+1+1
	OR: Copper is malleable, ductile and a good conductor (1); It is used in cables (1); Madhya Pradesh is the leading producer (1); Manganese is used in steel (1); Odisha is the leading producer (1).	1+1+1+1+1
20	Petroleum is found in the Mumbai High, Assam and Gujarat (1); It gives fuel and lubricants (1); Refineries process it (1); Natural gas is found with oil (1); It is used as CNG (1).	1+1+1+1+1
	OR: Iron ore is a ferrous mineral (1); Haematite and magnetite are main ores (1); Odisha, Jharkhand, Chhattisgarh and Karnataka have deposits (1); Ore is used in steel (1); Some is exported to Japan (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that **hematite** has more iron than magnetite; **magnetite** is richer (up to 70 per cent), though hematite is used more in quantity.
- Mixing the order of coal: peat, lignite, bituminous, **anthracite** (highest carbon, best quality).
- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** rock from which a metal can be extracted profitably	Source
2	**(a)** sunlight	Panels
3	**(a)** wind	Windmills
4	**(b)** steel making	Alloy
5	**(d)** processing used materials for reuse	Conservation
6	**(d)** Meghalaya	Narrow
7	**(b)** crude oil	Fossil fuel
8	**(a)** naturally occurring	Natural
9	**(b)** careful and sustainable use	Saving
10	**(d)** Compressed Natural Gas	Fuel

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mining is the extraction of minerals from the earth (1); It can be open-cast or underground (1).	1+1
12	Solar power is energy from sunlight (1); It is converted to electricity by photovoltaic cells (1).	1+1
13	A mineral is a naturally occurring substance with a definable internal structure (1); It is found in the earth's crust (1).	1+1
14	Wind power is energy from moving air (1); Windmills generate electricity (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Jharia (1); Raniganj (1); Bokaro (1).	1+1+1
16	Steel making (1); Machinery and tools (1); Construction (1).	1+1+1
17	Solar energy (1); Wind energy (1); Biogas and tidal energy (1).	1+1+1
	OR: Electric wires (1); Utensils (1); Electronic goods (1).	1+1+1
18	Aircraft (1); Electrical goods (1); Utensils and packaging (1).	1+1+1
	OR: Anthracite (1); Bituminous (1); Lignite or peat (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Use minerals judiciously (1); Recycle metals (1); Use substitutes (1); Save energy by switching off appliances (1); Use public transport (1).	1+1+1+1+1
	OR: Mica is a non-metallic insulator (1); It is used in electrical goods (1); Limestone is found in sedimentary rocks (1); It is the raw material of cement (1); It is also used in smelting iron (1).	1+1+1+1+1
20	Bauxite is the ore of aluminium (1); Aluminium is light and strong (1); Odisha is the leading producer (1); It is used in aircraft and utensils (1); It is a good conductor (1).	1+1+1+1+1
	OR: Thermal power burns coal, oil or gas (1); Plants are near coalfields (1); It meets much of India's demand (1); It causes pollution (1); Natural gas plants are cleaner (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that **hematite** has more iron than magnetite; **magnetite** is richer (up to 70 per cent), though hematite is used more in quantity.
- Mixing the order of coal: peat, lignite, bituminous, **anthracite** (highest carbon, best quality).
- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** uranium	Radioactive
2	**(c)** fossil fuel	Fuel
3	**(a)** oil	Assam
4	**(d)** iron ore	Rich in iron
5	**(b)** clean fossil fuel	CNG
6	**(d)** heat inside Earth	Hot springs
7	**(d)** electrical wires and cables	Conductor
8	**(c)** a low-grade brown coal	Soft
9	**(b)** water	Dams
10	**(c)** get exhausted with use	Finite

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	An ore is a mineral mixed with other substances from which metal is extracted (1); Iron ore is an example (1).	1+1
12	Petroleum is a liquid fossil fuel also called mineral oil (1); It is refined to get petrol, diesel and kerosene (1).	1+1
13	Coal is a solid fossil fuel formed from buried plants (1); It is used for power and industry (1).	1+1
14	Hydroelectricity is electricity from falling water (1); It is produced at dams (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fuel for vehicles (1); Lubricants (1); Raw material for petrochemicals (1).	1+1+1
16	Electrical insulators (1); Electronic goods (1); Paints and cosmetics (1).	1+1+1
17	Cement making (1); Smelting iron ore (1); Chemical industries (1).	1+1+1
	OR: Digboi (1); Mumbai High (1); Ankleshwar (1).	1+1+1
18	Bhakra Nangal (1); Hirakud (1); Tehri (1).	1+1+1
	OR: Tarapur (1); Kakrapara (1); Kalpakkam (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Wind energy is obtained by windmills (1); Tamil Nadu has the largest cluster (1); Biogas is made from dung and waste (1); It is used for cooking (1); Both are renewable (1).	1+1+1+1+1
	OR: Nuclear power uses uranium and thorium (1); Tarapur is a nuclear station (1); Solar energy uses sunlight (1); Photovoltaic cells produce electricity (1); Both are non-conventional (1).	1+1+1+1+1
20	Coal is the most abundant fossil fuel in India (1); Types are peat, lignite, bituminous, anthracite (1); Gondwana coal is in the Damodar valley (1); Neyveli has lignite (1); It is used for power and industry (1).	1+1+1+1+1
	OR: Hydroelectric power uses falling water (1); Dams store water for turbines (1); It is renewable and clean (1); Bhakra Nangal is an example (1); It needs suitable relief and water (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that **hematite** has more iron than magnetite; **magnetite** is richer (up to 70 per cent), though hematite is used more in quantity.
- Mixing the order of coal: peat, lignite, bituminous, **anthracite** (highest carbon, best quality).
- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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